Inputs and shared tools

PubChem ID check

Organic catalogue (CSV)
16 dyes: SMILES, InChIKey, PubChem ID, absorption band

Shared code + conda env
RDKit · PySCF · geomeTRIC
used by every step

Pipeline (molecules/), implemented and verified

1 Representation
SMILES → 3D (ETKDG + MMFF94s)
16/16 identifiers verified

2 DFT geometry optimisation
B3LYP/def2-SVP, ddCOSMO water
6 of 16 dyes optimised

MMFF (10)

DFT (6)

3 Frontier orbitals
HOMO, LUMO, gap; 16/16 converged
gap vs absorption: r = 0.89,
MAE 0.46 eV

4 Orbital character
isosurfaces, fragment populations
Mes-Acr+ donor–acceptor;
eosin Y dianion π→π*

5 Geometry check
MMFF vs DFT geometry
|Δgap| ≤ 0.03 eV rigid,
≤ 0.15 eV flexible

Outputs (results/molecules/)

CSV tables + figures

README (how to rerun)
REPORT (results, caveats)

Next steps (proposed order)

N1 DFT geometries for the other 10 dyes

N2 TDDFT excitation energies (S1, T1)

N3 def2-TZVP basis-set check

N4 Extend set: anthraquinone-2-sulfonate

implemented

planned

data file

external database